Investigating the Neural Correlates of Automaticity using EEG

Kayla Rail

1 Introduction

Picture a child learning how to ride a bike. When they first start, it is slow and effortful. Maintaining balance requires sustained attention, and trying to multitask (e.g., talk to their parent) is likely to topple them over. However, changes to the task environment, such as switching the locations of the front and rear brakes, is relatively easy to accommodate. With continued practice, these behavioural patterns switch places. Now they can happily multitask while riding smoothly, but are more prone to errors should the locations of the brakes change.

Riding a bike is an example of procedural learning. This is a type of skill (i.e., skill learning) where performance improves incrementally and leads to automaticity with continued practice. While this allows individuals to multitask with minimal effort, it is accompanied by impaired performance when task requirements change (Schneider and Shiffrin, 1977). Therefore, automatic behaviours are fast and efficient, but less adaptable to sudden changes in task structure. Multiple models of automaticity have emerged over the years, with its mechanism of action still under debate. My thesis aims to build upon and extend a prominent neurobiological theory of automaticity (Ashby et al., 2007) to account for its limitations which I will discuss shortly.

First, automaticity in skill learning is commonly studied using information-integration (II) category learning tasks (see Figure 1 for example category structure and stimuli). Here participants learn to assign stimuli to their respective categories incrementally via trial and error feedback, both hallmarks of procedural learning. Optimal performance requires the integration of multiple stimulus dimensions (e.g., bar angle and thickness) with the underlying categorisation rule being difficult to describe verbally (Ashby et al., 1998). As such, II category learning provides a well-established experimental model of procedural skill acquisition.

In 2007, Ashby and colleagues proposed a model of automaticity which suggests that automatic performance emerges as control is passed from subcortical to cortical pathways (Ashby et al., 2007; see Figure 2). It posits that initially, skill learning is driven by dopamine-dependent reinforcement-learning mechanisms within the basal ganglia. This means that the basal ganglia utilise dopamine as a teaching signal to reinforce correct responses or behaviours. Learning also occurs within cortex, however, this simple Hebbian mechanism (i.e., cells that fire together, wire together) does not possess the requisite teaching signals to reliably learn through trial and error. As such, the basal ganglia drives behaviour and 'teaches' cortex at the same time, utilising the simple Hebbian learning to strengthen the synpases associated with correct performance. With continued practice, the cortical pathway (indicated by the grey arrows in Figure 2) learns to perform tasks independently of subcortical input, meaning that the basal ganglia no longer contributes to performance, resulting in faster, more efficient behaviour.

Support for this theory comes primarily from longitudinal neuroimaging studies, which demonstrate a gradual shift in activation from the basal ganglia to cortical regions as learning progresses (Hélie et al., 2010; Waldschmidt and Ashby, 2011). Additional behavioural evidence and findings from special populations, such as individuals with Parkinson's disease (Ashby et al., 2003), further support the role of basal ganglia mechanisms in early procedural learning (for a review see Ashby and Valentin, 2018).

Despite this models success at accounting for a wide range of findings and behaviours, it's predictions about performance when task conditions change are inconsistent with recorded human behaviour (Cantwell et al., 2015; Maddox et al., 2010). During learning, participants gradually form stimulus-response (S-R) associations where a particular stimulus comes to trigger the response that has previously been rewarded.

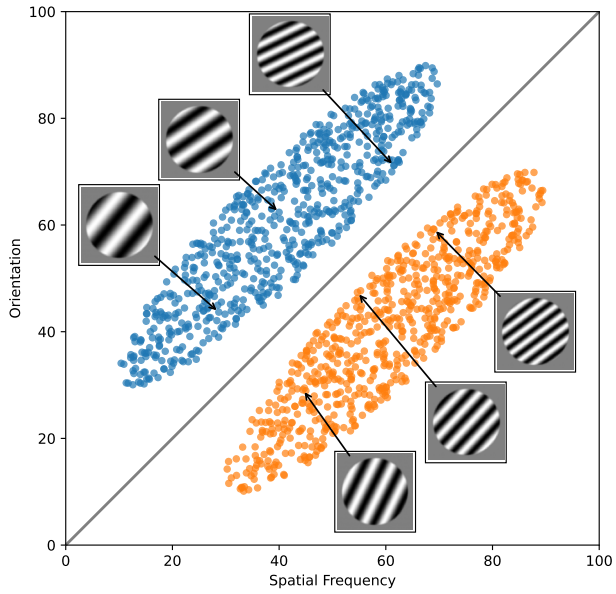

Figure 1: Information-Integration (II) stimulus category structure. Stimuli are circular sine-wave gratings with black and white lines that vary in orientation (angle) and thickness. Stimuli above the line (blue) are labelled Category A. Stimuli below (orange) are labelled Category B.

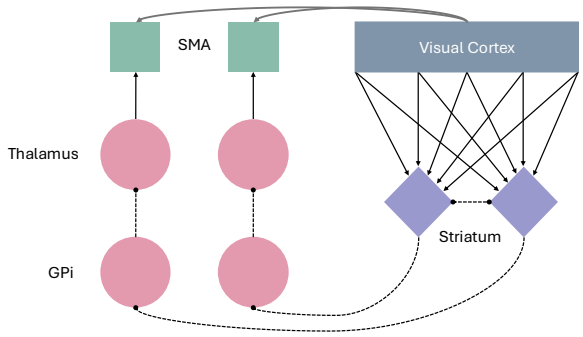

Figure 2: One-stage model of automaticity. Diamonds represent Izhikevich medium spiny neurons, circles represent quadratic integrate-and-fire units, and squares represent Izhikevich regular spiking neuron units. Solid lines with arrowheads represent excitatory input. Dashed lines with circular heads represent inhibitory input. Solid grey arrows indicate cortical-cortical connections. Dopamine-dependent reinforcement-learning synapse between the Substantia Nigra Pars Compacta and the Striatum is not shown. (GPi = Globus Pallidus internus, SMA = Supplementary Motor Area)

According to this model, if the learned category boundary (Figure 3A.) is rotated by 90 degrees (Figure 3B.) such that only some of the previously learned associations are incorrect, the models performance is moderately impacted. In contrast, when the boundary is rotated by 180 degrees (Figure 3C.) such that all previously learned S-R associations are now incorrect, the models performance is severely impaired and recovery is poor (Cantwell et al., 2015). However, human behaviour shows the opposite pattern with a larger performance cost following 90 degree rotations compared to 180 degrees (Maddox et al., 2010).

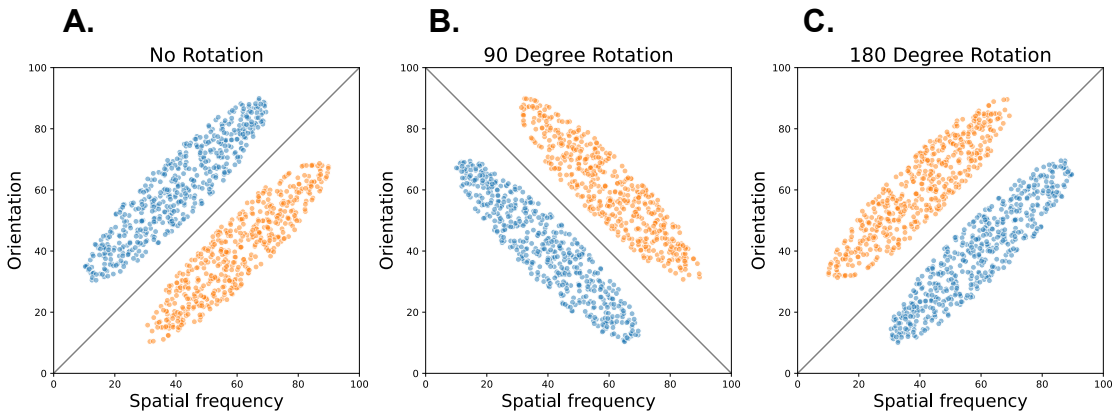

Figure 3: Learned and rotated II category structures. A. Learned category boundary (0 rotation). B. 90 degree rotation of category boundary. C. 180 degree rotation of category boundary. Blue coloured dots represent stimuli in Category A. Orange coloured dots represent stimuli in Category B.

A competing model was put forward by Cantwell, Crossley, and Ashby (Cantwell et al., 2015) who proposed a two-stage model of initial procedural learning (Figure 4). Here, learning occurs in two stages: the initial stage learns the perceptual category structure (e.g., this is Category A, this is Category B), and the second stage associates the category representations with motor responses (e.g., press the button on the left). This can better account for human behaviour when rotations occur. When the category boundary is rotated by 180 degrees, the category structure itself remains unchanged. Only the category representations associated with motor responses need to be updated which occurs in the second stage of learning. On the other hand, a 90 degree rotation alters the previously learned category structure, which requires relearning at both stages and results in greater performance impairment. Unlike the single-stage model, this account describes initial learning rather than the development of automaticity, as it does not include cortical-cortical pathways that would support automatic performance with extended practice.

My thesis aims to bridge this gap by creating and testing a two-stage model of automaticity. Specifically, we will test two competing implementations of this model. One in which visual cortex bypasses both stages of learning and projects directly to motor cortex (Figure 5A.), and another in which visual cortex projects to premotor cortex prior to motor output (Figure 5B.). In both models, initial learning is mediated by the two stage model that Cantwell et al. (2015) proposed but they differ in their cortical pathway. Additionally, the original double loop architecture has been simplified to reduce simulation time while preserving the functional properties relevant to the model. The remaining subcortical pathways maintain the same net excitatory input to their respective areas of cortex. These models make distinct predictions regarding the behavioural performance across different stages of learning which will be stated in the next section.

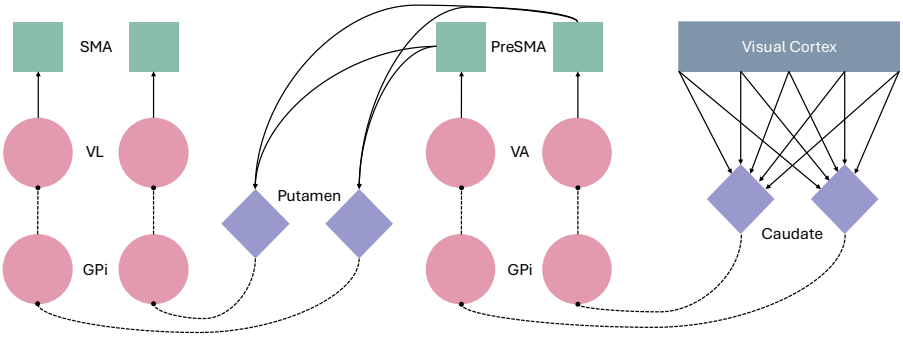

Figure 4: Two-Stage Model of initial learning. Diamonds represent Izhikevich medium spiny neurons, circles represent quadratic integrate-and-fire units, and squares represent Izhikevich regular spiking neuron units. Solid lines with arrowheads represent excitatory input. Dashed lines with circular heads represent inhibitory input. (GPi = Globus Pallidus internus, VA = ventral-anterior nucleus of the thalamus, VL = ventral-lateral nucleus of the thalamus, SMA = Supplementary Motor Area)

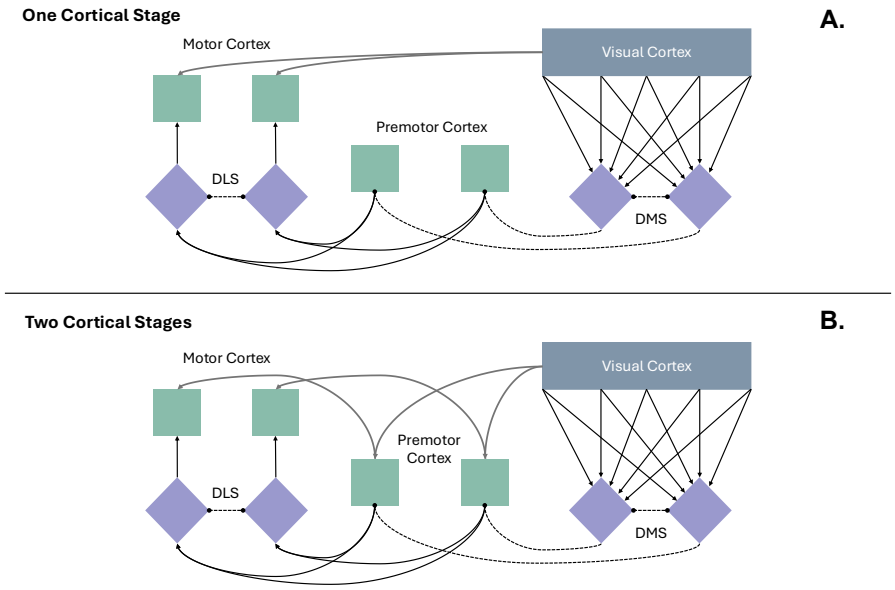

Figure 5: Proposed two-stage models of automaticity. Diamonds represent Izhikevich medium spiny neurons and squares represent Izhikevich regular spiking neuron units. Solid lines with arrowheads represent excitatory input. Dashed lines with circular heads represent inhibitory input. Solid grey arrows indicate cortical connections. (DMS = Dorsomedial Striatum, DLS = Dorsolateral Striatum)

In addition to behavioural measures, EEG provides a means of testing a key prediction shared by the current and newly proposed models of automaticity that have been difficult to examine using previous neuroimaging methods. Existing evidence suggests that activity gradually shifts from subcortical to cortical pathways with practice, but these approaches lack the temporal resolution required to detect differences in the timing of neural activity associated with these pathways. Specifically, signals travelling from visual cortex to motor output are expected to take longer through the basal ganglia, as there are more synapses, than the direct cortical pathway. Therefore, changes in the timing of cortical activity between the occipital and frontal lobes may provide an indication of which pathway (i.e., subcortical or cortical) is driving performance. To examine this, we will use EEG to track changes in functional connectivity, which is an estimation of how much activity in one cortical region is causing activity in another. According to the original single-stage model (Ashby et al., 2007), this measure of functional connectivity should increase with practice as control is passed from subcortical to cortical pathways.

To evaluate which of the models best accounts for automatic human behaviour, we will utilise a longitudinal training paradigm where participants will learn II categories over 21 days of practice. During this training period, EEG will be recorded in the lab on five days to examine changes in neural activity that accompany the development of automaticity. Also on these days, probe trials involving either 90 or 180 degree category boundary rotations will be used to explore the predictions of these competing models across early, intermediate, and late stages of learning.

This work contributes to broader theoretical understanding of how skills are acquired and automatised, and may inform translational research in skill training, ageing, and rehabilitation of maladaptive behaviours.

2 Research Aims and Hypotheses

The primary aim of this study is to evaluate competing models of automaticity in procedural category learning by examining behavioural and neural changes across extended practice.

Specifically, this study aims to:

1. Determine when learning transitions between early, intermediate, and automatic stages using behavioural responses to probe trials.
2. Evaluate whether behavioural performance following category-boundary rotations is better explained by a two-stage model of automaticity than by the existing single-stage model.
3. Examine how patterns of coordinated activity between visual and motor cortical regions change across training using EEG measures of functional connectivity.

We hypothesise that:

1. Participants will develop automaticity across 21 days of training, characterised by stable performance during dual-task conditions relative to single-task performance.
2. Under our first model of automaticity (Figure 5A.):
   - During early learning, the performance cost associated with 90 degree rotations will be greater than the cost associated with 180 degree rotations.
   - During intermediate stages of learning, the performance cost associated with both types of rotation will remain similar to that of early learning.
   - Following the development of automaticity, performance costs associated with both 90 and 180 degree rotations will significantly increase.
3. Under our second model of automaticity (Figure 5B.):
   - During early learning, the performance cost associated with 90 degree rotations will be greater than the cost associated with 180 degree rotations.

- During intermediate stages of learning, the performance cost associated with 90 degree rotations will significantly increase relative to the 180 degree rotation cost, which is expected to remain relatively stable.
- Following the development of automaticity, performance costs associated with 180 degree rotations will significantly increase and become comparable to those observed for 90 degree rotations.

4. Functional connectivity between posterior (visual) and anterior (motor) cortical regions will increase across training as performance becomes automatic

5. Neural responses to probe trials are expected to reflect increased decision uncertainty or surprise (however, given the exploratory nature of these measures, no directional hypotheses will be made between rotation conditions).

3 Methods

Participants. Participants will be approximately 20 undergraduate psychology PACE students recruited from Macquarie University. All participants will provide informed consent prior to participation.

Design and Procedure. Participants will learn II categories over 21 days of training. In this type of category learning task, participants view sine-wave gratings (for example stimuli see Figure 1) that vary in bar orientation (i.e., angle) and thickness. They then classify each stimulus into Category A or Category B using a keyboard response. Feedback indicating whether their category judgement response was correct or incorrect is presented after each trial, and they must use this to learn the category labels through trial and error (see Figure 6).

Although this task appears very simple and bears little resemblance to skills that eventually become automatic (e.g., riding a bike), there is strong evidence that the brain learns this task and automatises it using many of the same neural systems that are involved with more complex motor skills (Ashby and Valentin, 2018).

Participants will complete five sessions per week, consisting of one laboratory EEG session and four at-home training sessions. Laboratory sessions will consist of 550 standard training trials followed by 100 probe trials (650 trials total). During probe trials, the category boundary will be rotated by either 90 or 180 degrees (see Figure 3B. and 3C.). Participants will be randomly assigned to one of these probe conditions. At-home sessions will consist of 300 standard training trials and will not include probe trials. The testing schedule is summarised in Figure 7.

Following completion of training, participants will complete a dual-task session at home to assess the development of automaticity. During this session, participants will perform a working memory task concurrently with the categorisation task, similar to that used by Hélie et al. (2010). Automaticity is inferred if

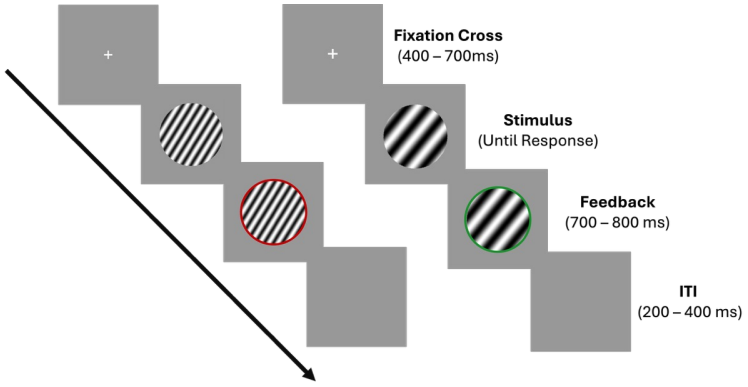

Figure 6: Information-Integration category learning task. Example trial sequence.

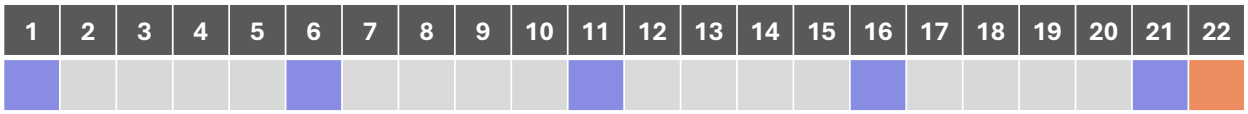

Figure 7: Overview of the 22-day experimental timeline for one participant. Each cell represents a testing day within the testing period. Purple boxes indicate EEG sessions (550 training plus 100 probe trials each). Grey boxes represent at-home training sessions (300 trials each). The orange box marks the dual-task session (300 trials). Total trials per participant = 8,350. Rest days are not shown.

performance remains stable under dual-task conditions, as automatic behaviours require fewer attentional resources and are therefore less disrupted by concurrent task demands.

For at-home sessions, the task will be administered using PsychoPy on participants' personal computers, and on laboratory computers during EEG sessions. Data will be uploaded to a secure university SharePoint system in accordance with institutional data-management protocols.

Behavioural Analysis. Behavioural performance will be assessed using accuracy and reaction time measures. The effect of probe trials will be quantified as performance cost, defined as the difference between performance immediately before probe trials and performance during probe trials. Changes in these costs across training will be used to evaluate predictions derived from the competing models of automaticity. Comparisons between training and dual-task performance will be used as an additional indicator of automaticity. All analyses will be conducted using Python.

EEG Recording and Analysis. EEG data will be recorded using a 64-channel BioSemi ActiveTwo system. Data will be pre-processed and analysed using MNE-Python. Standard preprocessing steps will include re-referencing, filtering, and artefact removal (e.g., eye blinks). Functional connectivity measures will be used to examine changes in coordinated activity between posterior and anterior cortical regions across training days.

Ethical Considerations. Ethical approval for the collection and storage of behavioural and electroencephalography (EEG) data has been obtained through the Macquarie University Human Research Ethics Committee (Project ID: 18907). Participation is voluntary and informed consent will be obtained prior to participation. Participants will be informed that involvement in the study will not influence their academic standing or PACE placement outcomes, and that they may withdraw at any time without penalty. These procedures are intended to minimise potential power differentials between the researcher and undergraduate participants.

EEG recording is non-invasive and presents minimal risk. Conductive gel will be used to ensure adequate signal quality, and standard laboratory hygiene and safety protocols will be followed.

Due to the technical requirements of EEG recording, some individuals may be unable to participate if adequate electrode contact with the scalp cannot be achieved (e.g., certain hairstyles or head coverings). This limitation will be communicated during recruitment and reflects methodological constraints rather than participant characteristics.

All data will be stored in de-identified form on secure university servers in accordance with Macquarie University data management policies. Risks associated with participation are minimal and primarily relate to fatigue from repeated task performance.

4 References

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5 Appendix

5.1 MRes Timeline

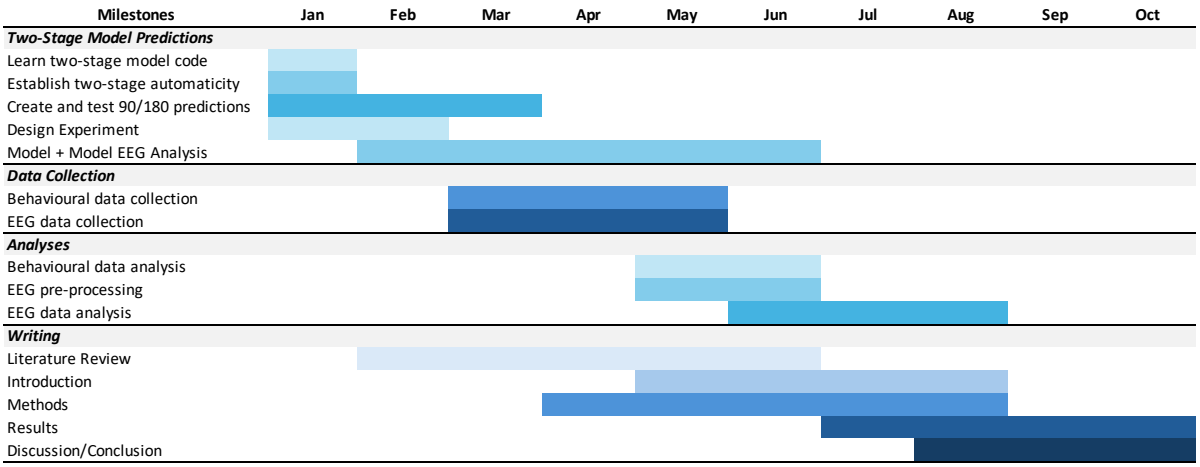

Figure 8: MRes Timeline

5.2 Budget

Item	Quantity	Unit Cost (AUD)	Subtotal (AUD)
BioSemi ActiveTwo/ActiveThree EEG Cap	1	$1,184.90	$1,184.90
EEG Signa Gel (250 g tubes)	12	$13.64	$163.80
Shipping Total			$90.95
Total Estimated Cost			**$1,439.65**

Table 1: Estimated equipment and consumables required for EEG data collection.

5.3 PhD Timeline

PhD Year 1: Retention and Modification of Automatic Skills

The first phase will focus on understanding how skills are retained once automaticity has developed, and whether established skills can be modified or "unlearned".

Retention of automatic skills

- Review literature on long-term retention of procedural and motor skills.
- Submit an ethics amendment to allow follow-up testing of participants who completed the automaticity study.
- Re-test participants approximately 6–12 months following initial training to assess retention of category knowledge and automatic performance.
- Use behavioural measures and EEG to examine whether neural markers of automaticity are maintained over time.

- Conduct behavioural and EEG analyses and prepare manuscripts for publication.

Modification and unlearning of automatic skills

- Extend existing work on skill modification, which has primarily focused on early learning rather than established automatic behaviour.
- Design an experiment in which participants first develop automaticity and are then exposed to manipulations intended to disrupt or modify learned categories (e.g., removal of veridical feedback).
- Examine whether behaviour reflects genuine unlearning or masking of previously learned associations.
- Check against current models and update predictions if required
- Obtain ethics approval, conduct data collection and analysis, and prepare manuscripts for publication.

Preparation for following year

- Submit ethics applications for stimulation protocols as they may take longer for approval.

PhD Year 2: Neural Modulation of Automaticity

The second phase will investigate whether the development of automaticity can be facilitated or disrupted through neural stimulation.

- Develop stimulation-based extensions of the automaticity paradigm using either peripheral nerve stimulation (PNS) or transcranial direct current stimulation (tDCS).
- Adapt the existing longitudinal training protocol such that laboratory sessions involve stimulation rather than EEG recording.
- Examine whether stimulation or inhibition alters the rate at which automaticity develops or changes behavioural sensitivity to task manipulations.
- Conduct data collection, analysis, and manuscript preparation.
- Evaluate where these findings sit amongst model predictions and update them if required.

PhD Year 3: Integration and Model Development

The final phase will integrate behavioural and neural findings across the PhD to refine theoretical accounts of automaticity.

- Integrate findings from retention, modification, and stimulation studies to evaluate competing models of automaticity.
- Extend computational modelling work to incorporate empirical findings from longitudinal behavioural and neural data.
- Conduct additional targeted experiments if required to resolve outstanding theoretical questions.
- Complete thesis write-up and prepare final publications.